

System Administration

Here, it's no different from system administration for Windows / UNIX / Mac. Although it comes under Linux, it is a broad area with a lot of options for super-specialisation. The system administrator's functional area is vast; he might have to perform the functions of computer maintenance, network administration, network security, platform migration, system analysis, and so on. A system administrator for Linux is responsible for installation, troubleshooting, and maintenance of IT hardware / software for the Linux platform by providing on-site administrative support to several departments in an organisation.

For the Linux platform, a good system administrator should necessarily have exhaustive knowledge and be capable of working cross-platform, that is, Linux, Windows, and UNIX variants.

Cluster servers consist of independent servers having different OSes to work together for higher

Being a freelance or consulting trainer in Linux is also a well-paying option

opt for specialisation in that environment on the Linux platform. Professionals already holding certifications from vendors like Oracle and SAP will have to acquire certifications of using database packages in cross-platform environments. The database vendors also offer modules for those already certified in Linux.

Web Server Administration

Candidates with a good knowledge of Web applications and Web hosting servers are suited best for this kind of job. The main job of the Web server admin is to maintain the Web site server(s) and hosted Web apps to ensure site responsiveness. Good understanding of dealing with HTTP requests and responses through the TCP/IP protocol suite is expected from Web server administrators. Here, a knowledge of basic programming and scripting is also required so as to be able to work in different server environments including Linux. Experience of working on any of the popular servers like Apache, MySQL, Microsoft SQL Server, Windows Server 2003, and Microsoft Exchange 2000 Server is necessary.

Training

The industry needs people with strong Linux basics to train others working with it. Trainers are hired for corporate training, educational institutions, and governments. Organisations that offer training services as their core business operation are major employers of Linux trainers. Training is a well-paying option.

Being a freelancer or consultant for any of the above-mentioned career paths is also a well-paying option. Organisations shifting to Linux consult Linux specialists for specific purposes, such as installing a Linux server for their day-to-day operations. Yet, like freelancing options in other fields, it depends upon how one has networked and is recognised amongst peer groups.

Remuneration And The Growth Path

Organisations across industries follow their own policies in assigning designations and promotions, and offering remuneration. The functional areas in Linux can be divided into the top, middle, and bottom levels. Designations like "Junior Programmer," "Assistant System Administrator," etc. are assigned to Linux professionals. In organisations that have employees for both the Linux and Windows platforms, the platform tag is carried separately after the designation.

The average time taken to grow level-wise is approximately three to five years for any functional area. And like in any other occupation, salaries are offered based on qualification, performance, and project contribution. (Refer Table Remuneration)

In terms of hard skills, organisations look for skills specific to the employee's functional domain. So for application developers, what matters is knowledge of tools, programming languages, any applications developed, period since the candidate has been developing programs, etc. Soft skills are invariably important. Hands-on working ability with computer

Approximate Remuneration Matrix

Department	Entry Level	Middle Level	Top Level
System Administration / Network Administration / Web Server Administration	Rs 15,000 - 35,000	Rs 35,000 - 50,000	Rs 60,000 - 83,000
Tech Support / Desktop Management and Maintenance	Rs 8,000 - 16,000	Rs 18,000 - 35,000	Rs 40,000 - 60,000
Device Drivers for Embedded Systems and Kernel Development	Rs 15,000 - 35,000	Rs 35,000 - 60,000	Rs 70,000 - 1,00,000
Applications Development	Rs 12,000 - 30,000	Rs 35,000 - 50,000	Rs 60,000 - 75,000

All figures mentioned above are per-month, aggregated, and indicative.

performance. With cluster servers and virtualisation technology becoming popular, administrators are required to manage these as well. Demand for such professionals is expected to shoot up as many enterprises shift to Linux.

Network Administrators And Security Engineers

Network administrators are responsible for installation, configuration, and analysis of computer networks in an organisation. Their daily activities include troubleshooting of network problems, reducing network bottlenecks, monitoring the network, and securing it. A Linux network administrator plays a crucial role in any "Linux-enabled" organisation.

But Linux network administrators are essentially required for security of data at data centres. Confidential information is safeguarded with regular backups, ensuring secured connections, etc. Network security engineers work upon testing, configuring firewalls, and monitoring data flow. Hacking can be made next to impossible if proper security mechanisms are employed by the network security engineer.

Database Administration

Database professionals who hold certifications of a particular database environment may well